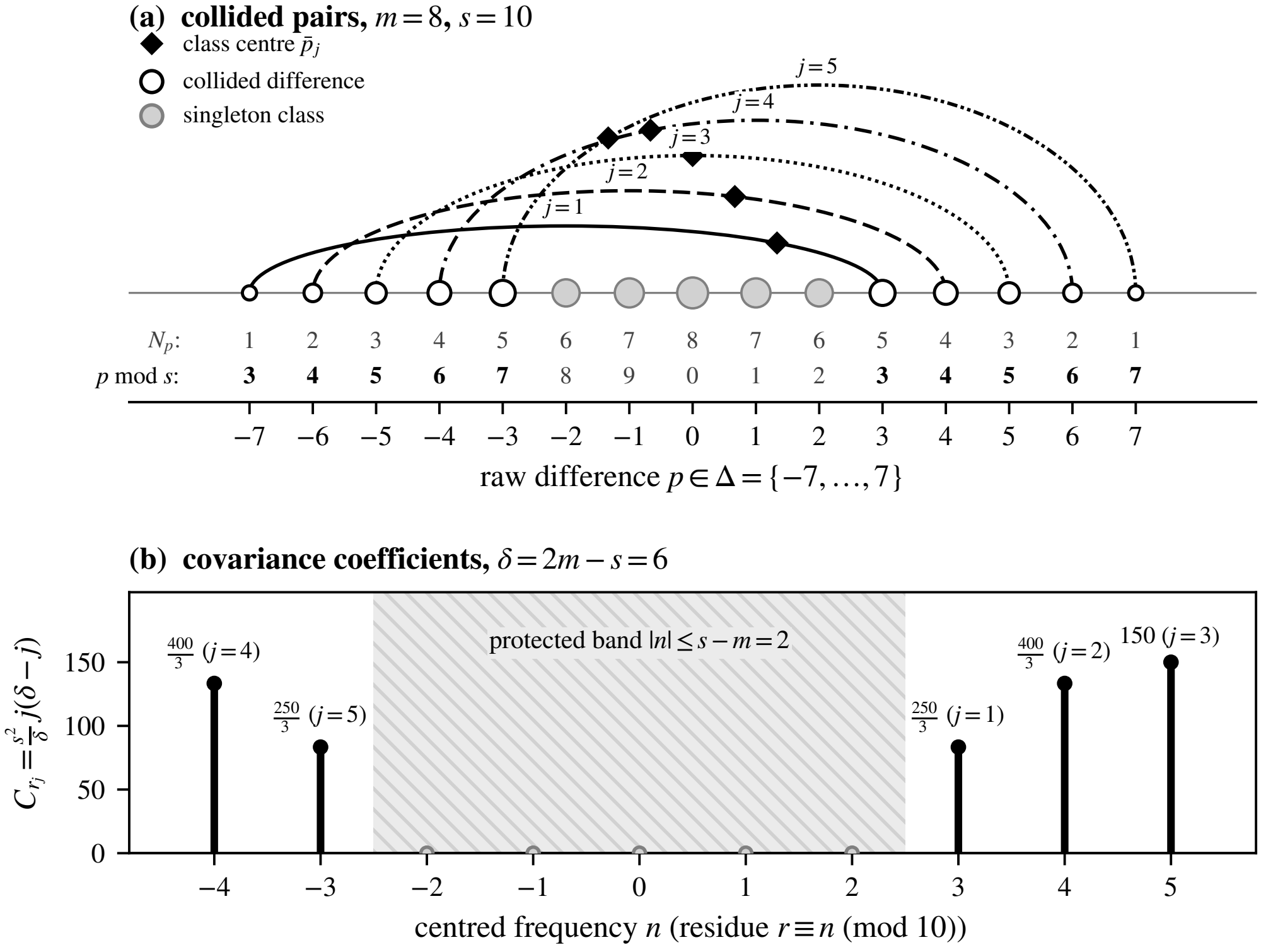
(a) collided pairs, $m = 8$, $s = 10$
class centre $\bar{p}_j$
collided difference
singleton class
$j = 5$
$j = 4$
$j = 3$
$j = 2$
$j = 1$
N_p: 1 2 3 4 5 6 7 8 7 6 5 4 3 2 1
p mod s: 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7
−7 −6 −5 −4 −3 −2 −1 0 1 2 3 4 5 6 7
raw difference $p \in \Delta = \{-7, \ldots, 7\}$
(b) covariance coefficients, $\delta = 2m - s = 6$
$C_{r_j} = \frac{s^2}{\delta} j(\delta - j)$
protected band $|n| \le s - m = 2$
$\frac{400}{3}$ $(j = 4)$
$\frac{250}{3}$ $(j = 5)$
$\frac{250}{3}$ $(j = 1)$
$\frac{400}{3}$ $(j = 2)$
150 $(j = 3)$
0 50 100 150
−4 −3 −2 −1 0 1 2 3 4 5
centred frequency n (residue $r \equiv n$ (mod 10))